Rohan Shenoy

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Education

Degree	Institute/Board	GPA/Percentage	Year(s)
Graduate Student	Caltech	-	08/2023-*present*
-	Starting PhD process	-	06/2023-08/2023
Bachelor of Science	University of California, San Diego	3.98	09/2019-06/2023
11 - 12th Grade	Indian School Certificate (ISC)	94%	2017-19
9 -10th Grade	Indian Certificate of Secondary Education (ICSE)	96.6%	2015-17

Honors and Awards

- **David and Ellen Lee Distinguished Fellow**,"[A]warded to [Caltech's] most promising students to allow them freedom to focus totally on their research." $ 46,500
- **TRELS Scholarship**,For improving Autoencoder performance in the High Granularity Calorimeter $ 1000 *06/2021*
- **Undergraduate Summer Research Award**,For developing graph neural networks with object condensation loss for the High Granularity Calorimeter Level 1 Trigger $ 4000 *06/2022*
- **UC San Diego Chairs Challenge Award**,Travel award to present research $ 500 *01/2022*

Publications and Presentations

- **Radiation Impact on Superconducting Qubits (RISQ) Workshop**,
  Simulation of low-temperature blackbody radiation backgrounds — poster *06/2026*
- **Machine Learning: Science and Technology**,
  Differentiable Earth mover's distance for data compression at the high-luminosity LHC *2023*
- **A3D3: Accelerated Artificial Intelligence Algorithms for Data-Driven Discovery Annual Meeting**,
  Deep Learning for the High Granularity Calorimeter **(HGCAL)** *2022*
- **Fast Machine Learning for Science Workshop 2022**,
  EMD Neural Network Loss for the ECON-T ASIC Autoencoder *2022*
- **Undergraduate Summer Research Poster Session**,
  Deep Learning for the High Granularity Calorimeter L1Trigger *2022*
- **American Physical Society, April Meeting**,
  EMD Neural Network for HGCAL Data Compression Encoder ASIC *2022*

Projects

- **Background modeling and limit setting for SuperCDMS SNOLAB** *2024 - present*
  Advisor: Sunil Golwala, PhD
  - Building a flexible Python tool to set constraints on low-mass dark matter interactions.
  - Extended maximum-likelihood fit comparing signal against the full background model: tritium, neutrons, coherent neutrino scattering, Compton events, surface betas, radon exposure and in-situ cosmic rays.
  - Detector resolution and energy yield carried as fitted nuisance parameters rather than fixed assumptions; fiducial volume cuts applied.
  - Validation against Monte Carlo trial datasets generated across the parameter space.
  - Passed PhD candidacy on this work, Fall 2025.

- **Simulation of low-temperature blackbody radiation backgrounds** *2024 - present*
  Advisor: Sunil Golwala, PhD
  - Extending GEANT4 to propagate blackbody photons emitted by warm fridge surfaces, where ray-like optical propagation fails because photon wavelengths approach the scale of gaps between detector components.
  - Scripted ANSYS HFSS in Python to solve for transmission coefficients and near- and far-field electric fields across waveguide geometries.
  - Converted field solutions into sampled transmission probabilities and angular emission profiles for the simulation.
  - Diffraction implemented and validated for a single photon frequency and crack dimension; presented as a poster at the RISQ workshop.
  - Applies to both SuperCDMS sensor noise and pair-breaking backgrounds in superconducting qubits.

- **Graph neural networks and object condensation for the HGCAL ASIC latent space** *June 2022 - 2023*
  Supervisor: Javier Duarte, PhD **Github**
  – Particle reconstruction from the HGCAL latent space.
  – Used Pytorch Geometric to develop a GNN for this task.
  – Achieved pileup mitigation with 85 % accuracy in high luminosity-LHC 200 pileup events.
  – Presented at A3D3: Accelerated Artificial Intelligence Algorithms for Data-Driven Discovery Annual Meeting.

- **Leptoquarks for the $R(D^*)$ anomaly** *March 2022 - 2023*
  Supervisor: Benjamin Grinstein, PhD **Project Overview**
  – Explored scalar leptoquarks as a possible solution for the excess in the B meson to D+τ transition.
  – Search was conducted in pp $\rightarrow$ b + c + τ + MET final state at the LHC.
  – Developed scalar LQ models with various parameters using FeynRules.
  – Used MadGraph_aMCNLO as a MC simulation tool.
  – Checked results at leading order analytically.
  – Demonstrated enhanced sensitivity in the $m_T(\tau)$ tail compared to previous searches.

- **Improved autoencoder training for HGCAL Level 1 Trigger** *July 2021 - 2023*
  Supervisors: Javier Duarte, Jim Hirschauer, Danny Noonan, Cristina Ana Mantilla-Suarez, Nhan Tran **Github**
  – Working in collaboration with Fermilab to improve the on-detector ASIC encoder.
  – Event level cuts of Monte Carlo data using uproot and awkward.
  – Learnt basics of CMSSW for L1Trigger simulation.
  – Presented at 2022 Fast Machine Learning for Science Workshop.

- **Differentiable approximation to the Earth mover's distance** *March 2021 - 2023*
  Principal Investigator: Javier Duarte, PhD **Github**
  – Developed a differentiable, fast approximation to the Earth mover's distance.
  – Employed deep learning techniques known as convolutional neural networks written in tensorflow.
  – Used this metric to achieve better physics results for front-end data compression at the High Luminosity Large Hadron Collider.
  – Work was selected for presentation at the April 2022 Meeting of the American Physical Society.
  – Paper in pre-print.

Abstracts

- **Deep Learning for the High Granularity Calorimeter** *October 2022*
  Presented at NSF A3D3 Institute Annual Meeting and UCSD Undergraduate Poster Session
  The High Granularity Calorimeter (HGCAL) is part of the High Luminosity upgrade of the CMS detector at the Large Hadron Collider (HL-LHC). This upgrade introduces the issue of large amounts of data being generated at the front-end of the detector. One resolution is to use deep-learning techniques for real time data compression and event selection. For the event selection technique, also known as a "trigger", the HGCAL has over 6 million channels. The trigger uses a neural network embedded on Application Specific Integrated Circuits (ASICs). The ASIC foresees an encoder based on a convolutional neural network (CNN). The performance is evaluated using the earth mover's distance (EMD). Ideally, we would like to quantify the loss between the input and the decoded image at every step of the training using the EMD. However, the EMD is not differentiable and can therefore not be used directly as a loss function for gradient descent. We use a convolutional neural network to approximate the metric and make it differentiable for gradient descent in the autoencoder. The optimal latent space produced with this approach to data compression contains information about the original particle physics event. **We first represent this latent space as a graph, and then use a convolutional graph neural network to learn the physics of the latent space.**

- **EMD Neural Network Loss for the ECON-T ASIC Autoencoder** *April, October 2022*
  Presented at Fast Machine Learning for Science Workshop 2022, and American Physical Society April Meeting
  The High Granularity Calorimeter (HGCAL) is part of the High Luminosity upgrade of the CMS detector at the Large Hadron Collider (HL-LHC). For the trigger primitive generation of the 6 million channels in this detector, data compression at the front end may be accomplished by using deep-learning techniques using an on-ASICs network. The Endcap Trigger Concentrator (ECON-T) ASIC foresees an encoder based on a convolutional neural network (CNN). The performance is evaluated using the earth mover's distance (EMD). Ideally, we would like to quantify the loss between the input and the decoded image at every step of the training using the EMD. However, the EMD is not differentiable and can therefore not be used directly as a loss function for gradient descent. The task of this project is to approximate the EMD using a separate set of CNNs and then implement the EMD NN as a custom loss for the ASIC encoder training, with the goal of achieving better physics performance.

Experience

- **Golwala Lab** *2023 - present*
  Graduate Student Researcher Caltech
  – Member of the SuperCDMS SNOLAB collaboration, searching for low-mass dark matter with cryogenic germanium and silicon detectors operated at 15 mK, two kilometres underground.
  – Statistical background modeling and limit setting; simulation of blackbody radiation backgrounds; monitoring of the running experiment.
  – David and Ellen Lee Distinguished Fellow. Passed PhD candidacy Fall 2025.

- **Duarte Lab** *2021 - 2023*
  Undergraduate Researcher UCSD
  – Worked at the intersection of Particle Physics and Artificial Intelligence Techniques. Professor Duarte's Lab is part of the CMS collaboration at the Large Hadron Collider and uses geometric deep learning techniques to learn more about the Higgs Boson and aid in the search for exotic new physics.

- **High energy phenomenology** *2022 - 2023*
  Undergraduate Researcher UCSD
  – Worked with Professor Benjamin Grinstein on a phenomenological study of leptoquarks at the LHC.

Technical Skills

- **Statistics and Machine Learning**: Extended maximum-likelihood fitting, nuisance-parameter treatment of systematics, Monte Carlo methods, limit setting (Minuit2). PyTorch, PyTorch Geometric, TensorFlow: CNNs, graph neural networks, autoencoders, custom differentiable losses.
- **Simulation**: GEANT4, ANSYS HFSS (scripted in Python), MadGraph_aMC@NLO, MadAnalysis, FeynRules.
- **High Energy Physics**: Experiment: CMSSW, CMS Open Data, uproot, awkward, coffea.
  Phenomenology: basics of Feynman diagrams and scattering amplitudes.
- **Programming Languages**: Python, Java, MATLAB, Mathematica.
- **Other**: LaTeX, Github, Gitlab.

Key courses taken

- **Physics**: Quantum Field Theory, Atomic Physics, Graduate Quantum Mechanics, Advanced Circuits Lab, Mathematical Methods, Nuclear Physics. Honors Physics First 2 Years.
- **Mathematics**: Honors Linear Algebra, Advanced Linear Algebra, Modern Algebra/Group Theory, Complex Analysis, Differential Equations.

Positions of Responsibility

- **Vice President**, CGPU-UAW Local 2478 — Caltech's first labor union, 2,000 members represented *2025-2026*
- **Treasurer**, Caltech Graduate Student Council — $120,000 budget, closed at zero deficit *2024-2025*
- **Member**, Society for Physics Students, ARMS(Advocating for and Representing Minority Students) *2021-2023*
- **Board Member**, Groundwork Books *2019-Present*
- **Prefect**, Smt. Sulochanadevi Singhania School *2017*

Volunteer experience

- **Groundwork Books Collective** *2019 - Present*
  Board Member UCSD
  – Groundwork Books Collective is a student-run workers cooperative.
  – 1000 hours of volunteer work at a non-profit bookstore.
  – Organizing and outreach: established connections with various social justice organizations, to host events and community education.
  – Restock: restarted inventory purchasing for the first time since 2009; sold political theory, fiction, used books and cheap iClickers to undergraduate students.
  – Recruited and trained volunteer staff, and handled the cooperative's tax filings.